Which chatbot generated the most racial and ethnic stereotypes?

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Despite the growing popularity of publicly available language models (Urchs et al. 2023) and the increasing trust we place in AI, even in the context of academic activities, it is still important to emphasize its imperfection. They are mere "stochastic parrots" that repeat and assemble into seemingly new wholes training data that is non-ideal and outdated (Zhou, Sanfilippo 2023), in addition, collected on a massive scale that cannot be subjected to any form of validation, either linguistically, or in terms of content (Salinas, Burbat, 2023).

This proposal presents a comparative study of three of the most popular AI systems in terms of the inclusion of hurtful gender and ethnic/racial stereotypes in the texts they generate. That such stereotypes occur is not in doubt (Duncan, McCulloh 2023). The occurrence of hurtful gender stereotypes in texts may be related, among other things, to the underrepresentation of certain groups in texts used in training (Bolukbasi et al. 2016), or at least to the need to translate messages between languages that overtly encode the gender category in language (Vanmassenhove, Hardmeier and Way 2019; Rescigno, Monti 2023), or by having to include or being able to omit personal pronouns in the message (Zou, Schiebinger 2018). Unfortunately, there is still little of this type of research done on linguistic material found in the context of hurtful ethnic and racial stereotypes (Hoffmann et al., 2024). That ethnic and racial stereotypes occur in texts automatically generated by ChatGPT was demonstrated in a talk at the Digital Humanities 2024 conference last year (Rykowska, 2024).

This poster, therefore, aims to explore in more depth the racial and ethnic stereotypes occurring in texts automatically generated by the three most popular language models, ChatGPT, Google Bard and Claude AI. The study covers automatically generated stories in three categories: one with an unspecified central theme, a happy ending one, and a story in which the main character fails. The main character is a representative of some minority. In addition to the linguistic layer of the texts, the analysis also takes into account the titles that the chatbot gives to the stories, the names that are given to the characters in the stories, and the gender that is chosen for each of the main characters (depending on the theme and the positive/negative ending of the story).

Due to the fact that ChatGPT, when asked to select racial and ethnic minorities, is characterized by stereotyping already at this stage of the study, 25 minority groups were arbitrarily selected[1]. The study was conducted in four languages: English, French, Polish, and Lithuanian. Thus, these languages represent four different language families, two of which are characterized by their analyticity and two by their inflectionality. Additionally, English and French are considered to be well-resourced languages in contrast to Polish and Lithuanian, which are underrepresented in the training data.

The generated texts were first divided by the gender of the main character. They were further analyzed using keyword extraction tools, taking into account the opposition of the two corpora. The analysis was carried out using the oppose() function in stylo package for R (Eder, Rybicki, Kestemont, 2016). In addition, a topic modeling analysis was also carried out for both corpora in order to extract story themes typical for stories where the main character is female and also for those where the main character is male.

In all the texts analyzed, words relating to artistic skills, especially handicrafts and the cultivation of traditions, were far more frequent in texts where the main character was female. Stories where the main character was a man were characterized by words describing conflicts, including armed conflicts and fighting for the welfare of the whole community. In addition, the analysis of the names of the main characters showed interesting results, which were often just very randomly chosen words from minority languages, although there were also cases of hidden stereotypes in the names, as in the case of the main character coming from an Apache group called by ChatGPT Rojo 'red'.

A comparison of the hidden stereotypes with their quantitative analysis will be presented in the poster[2]. The analysis shows that most hidden stereotypes are found in the texts generated by ChatGPT-3.5, but there is no shortage of them in the other two AI systems as well.

Notes

1. The list of groups and the generated texts were uploaded to a remote repository on Github (https://github.com/aleksandra2812/EthnoBias).
2. The results of the analysis and all graphs can also be found in the repository on Github linked above.

Bibliography

Bolukbasi Tolga et al. (2016). Man is to computer programmer as woman is to homemaker? Quantifying and Reducing Stereotypes in Word Embeddings. To appear in Advances in Neural Information Processing Systems (NIPS).

Clay D. / McCulloh I. (2023), Unmasking Bias in Chat GPT Responses, *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining* , pp. 687-691.

Eder, M. / Rybicki, J. / Kestemont, M. (2016). Stylometry with R: a package for computational text analysis. *R Journal* 8(1): 107-121. https://journal.r-project.org/archive/2016/RJ-2016-007/index.html

Hoffman, V. et al. (2024). AI generates covertly racist decisions about people based on their dialect. *Nature* 633: 147–154.

Rescigno, A. A. / Monti, J. (2023): Gender Bias in Machine Translation: a statistical evaluation of Google Translate and DeepL for English, Italian and German. In *Proceedings of the International Conference HiT-IT* 2, 1-11.

Rykowska, A. (2024). Gender and ethnic bias in machine translation and AI-generated texts. Presentation in *Digital Humanities 2024* , Washington DC.

Salinas, M. J. V. / Burbat, R. (2023). Google Translate and DeepL: breaking taboos in translator training. Observational study and analysis, *Ibérica* 45: 243-266.

Urchs, S. et al. (2023), How Prevalent is Gender Bias in ChatGPT? – Exploring German and English ChatGPT Responses, *1st Workshop on Biased Data in Conversational Agents* , arXiv:2310.03031.

Vanmassenhove, E. et al. (2019): Getting Gender Right in Neural Machine Translation. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing* . Association for Computational Linguistics, 3003-3008.

Zou, J. / Schiebinger, L. (2018). Design AI so that it's fair. *Nature* 559, 324-326.

Zhou K. Z. / Sanfilippo M. R. (2023). Public Perceptions of Gender Bias in Large Language Models: Case of ChatGPT and Ernie, pp. 1-10, arXiv:2309.09120v1.